Feature importance of the 15:30 forecast (16:00 bar)

MDI, split count, permutation importance on the causal held-out tail, SHAP — trees and the per-bar linear models; 1,469 forecasts 2018-06-25 .. 2024-04-30, 147 refits

Setup. Models: per-bar ridge and lasso (coefficients re-solved every session, penalty re-chosen causally every 250 sessions) and the untuned per-bar LightGBM, XGBoost and random forest (refit every 10 sessions) — the shipped configurations. Every fit uses the 2000 sessions before it. A refit's *held-out tail* is the $\leq$ 10 sessions its model forecasts, up to the next refit: never in its fit, so a loss change measured there is out of sample and causal. Buckets: all features, and HAR + calendar. Library: scikit-learn 1.9.0 (random forest), LightGBM 4.6.0, XGBoost 3.2.0, shap 0.51.0; the permutation and drop-column loops are written out, because a library permutation routine scores one fitted model on one test set, not 147 walk-forward models on their own tails.

Measures. (1) *MDI / gain*: LightGBM gain, XGBoost total gain, random-forest impurity decrease; share per refit, averaged. (2) *Split count*. (3) *Permutation importance* on the tail, loss = QLIKE of the plain back-transform $\hat{y}^2\times$ diurnal baseline against the bar's realized variance (no recalibration), and squared error in the fit space; 10 draws per unit and refit, the same draws for all five models. P1 shuffles the unit's values among the tail's rows (the literal permutation; blind to slow inputs, which barely move in 10 sessions); P2 replaces them with rows drawn from the refit's own training window (past data only). A unit is a column, a *series* (all columns of one input: lags, flags, gates) or a *cluster* (columns every pair of which has $|\text{corr}| \geq 0.8$, complete linkage); a group's columns move together. Intervals: 95% circular-block bootstrap over refits (block 6). (4) *SHAP*: TreeSHAP (path-dependent) on the same tail rows; for the linear models $\beta_j(x_j - \bar{x}_j)$ with the window mean as background. Linear extras: $|\beta \cdot sd|$ and drop-column importance (re-solve without the unit, penalty held).

Key. Every exogenous input is divided by its time-of-day baseline, and the HAR columns `har_ma_*` are trailing means of the adjusted target, so no name carries a prefix. `har_ma_*` realized variance of the bar (the target's own HAR ladder, means over the last 1, 5, 25, 125, 625, 3125 bars; `har_ma_*_x_close` = the same times the 16:00 close gate, identical at this bar); `sumabsret_ma_*` absolute return; `sumret_ma_*` signed return; `sumret3/4_ma_*` 3rd/4th-power returns; `sumpret2_ma_*` upside squared returns; `sumbipow_ma_*` bipower variation; `sumautocov_ma_*` return autocovariance; `sumvolume_ma_*` ES volume; `numobs_ma_*` ES prints per bar; `vix/vvix/vix3m_ma_*` the Cboe indices; `fomc_*` FOMC flags and distances; `*_avail/active_ma_*` is-present / is-nonzero flags; calendar = `DOW_*`, `is_*`, `hour`, `days_to_opex`. all features adds the constituent cross-section (`*_ewstock`, `*_vwstock`), Cboe volume (`voldemand_*`) and StockTwits (`stocktwits_*`).

Table 1: The design is float throughout, but not all continuous: distinct values per column in each 2000-session window (median over the 147 windows) — what MDI's cardinality bias keys on.

bucket	columns	constant	binary	3–25	26–500	>500	series	clusters (columns)
HAR + calendar	34	10	11	0	1	12	2	5 (14)
all features	640	143	40	93	91	273	49	101 (505)

1. Does MDI's cardinality bias matter here? Yes, below the leader.

Table 2: Spearman rank correlation, over the non-constant columns, between a column's importance and its number of distinct values. MDI, split count and TreeSHAP track cardinality; permutation importance does not.

bucket	model	MDI	split	SHAP	perm P1 QL	perm P2 QL	perm P2 MSE
all features	LightGBM	0.84	0.84	0.81	0.02	-0.06	-0.03
all features	XGBoost	0.83	0.83	0.81	0.19	0.17	0.06
all features	random forest	0.90	0.91	0.88	0.08	0.05	-0.02
HAR + calendar	LightGBM	0.85	0.84	0.72	0.23	0.20	0.04
HAR + calendar	XGBoost	0.79	0.79	0.73	0.33	0.43	0.41
HAR + calendar	random forest	0.86	0.83	0.80	0.11	0.38	0.57

Table 3: Noise probes: on every third refit a second fit appends three pure-noise columns (continuous; 20 values; binary). Median rank among all columns (1 = most important) and, in brackets, the share of the real columns the model uses that rank below the probe; permutation P2 of the continuous probe (ΔQLIKE $\times 10^3$, 95% interval) against `har_ma_*`'s.

bucket	model	cols	MDI rank	split rank	SHAP rank	20-val MDI	binary MDI	perm P2 (cont.)	`har_ma_*`
all features	LightGBM	643	77 (75%)	47 (83%)	100 (69%)	153	319	−0.08 [-0.33, +0.14]	294
all features	XGBoost	643	117 (64%)	80 (72%)	131 (60%)	166	319	+0.17 [+0.01, +0.38]	280
all features	random forest	643	48 (89%)	10 (98%)	92 (79%)	108	317	−0.04 [-0.40, +0.33]	368
HAR + calendar	LightGBM	37	8 (72%)	5 (84%)	11 (60%)	11	16	+0.08 [-1.51, +1.81]	630
HAR + calendar	XGBoost	37	9 (64%)	6 (80%)	13 (51%)	13	19	+0.50 [-0.22, +1.17]	588
HAR + calendar	random forest	37	5 (80%)	1 (100%)	11 (60%)	9	15	−0.35 [-2.26, +1.60]	691

Reading. A column of pure noise with a new value on every row is ranked by MDI above 75%, 64%, 89% (LightGBM, XGBoost, forest) of the real all-features columns the model uses, and by the forest's split count above 98%; the binary noise column sits near the bottom. The same noise gets a permutation importance of at most 0.50×10^{-3} in absolute value (5 of 6 intervals include zero; the exceptions all features XGBoost [+0.01, +0.38]e-3), against at least 0.28 for `har_ma_*`. TreeSHAP, which splits the fitted trees, inherits part of the bias. Over the real columns, MDI, split count and SHAP correlate 0.83–0.88 with the number of distinct values, permutation -0.06–0.19. So below the leader, the MDI / split ranking of this design is partly a ranking by cardinality; the permutation ranking is the one to use for selection. At the series level the bias does not reach the top: every tree measure puts `har_ma_*` first. Series that MDI or split count puts in the top 5 but permutation P2 ranks outside the top 10 (permutation rank in brackets): all features LightGBM: `sellturnover_vwstock_ma_*` (35), `stocktwits_sentiment_ma_*` (49), `sumret3_ma_*` (28); all features XGBoost: `sumautocov_ma_*` (11); all features random forest: `voldemand_spx_open_and_close_ma_*` (47), `voldemand_all_open_only_ma_*` (36), `voldemand_spx_open_only_ma_*` (43).

2. The top inputs by measure

Table 4: All-features design, series level: top three per model and measure (share for MDI / split / $|\beta \cdot sd|$ / SHAP; change as % of the model's tail loss for permutation and drop-column).

model	measure	top three
ridge	$\|\beta \cdot sd\|$	`har_ma_*` 11%; `sumabsret_ma_*` 5%; `sumabsret_vwstock_ma_*` 5%
ridge	perm. P1, ΔMSE	`har_ma_*` +63%; `sumabsret_ma_*` +7%; `sumabsret_vwstock_ma_*` +7%
ridge	perm. P2, ΔMSE	`har_ma_*` +142%; `sumabsret_ma_*` +22%; `sumabsret_vwstock_ma_*` +18%
ridge	perm. P2, ΔQLIKE	`sumautocov_ma_*` +4051120%; `spread_vwstock_ma_*` +57340%; `voldemand_spx_open_only_ma_*` +3749%
ridge	drop-column, ΔMSE	`har_ma_*` +17%; `calendar` +2%; `sumvolume_ma_*` +2%
ridge	SHAP mean $\|\phi\|$	`har_ma_*` 15%; `sumabsret_ma_*` 6%; `sumabsret_vwstock_ma_*` 5%
lasso	$\|\beta \cdot sd\|$	`har_ma_*` 46%; `sumabsret_vwstock_ma_*` 10%; `sumret_ma_*` 9%
lasso	perm. P1, ΔMSE	`har_ma_*` +145%; `sumabsret_vwstock_ma_*` +8%; `sumret_ma_*` +7%
lasso	perm. P2, ΔMSE	`har_ma_*` +290%; `sumabsret_vwstock_ma_*` +18%; `vix_ma_*` +15%
lasso	perm. P2, ΔQLIKE	`har_ma_*` +259%; `vix3m_ma_*` +85%; `sumbipow_ewstock_ma_*` +48%
lasso	drop-column, ΔMSE	`har_ma_*` +27%; `sumret_ma_*` +2%; `sumabsret_vwstock_ma_*` +1%
lasso	SHAP mean $\|\phi\|$	`har_ma_*` 46%; `sumabsret_vwstock_ma_*` 10%; `sumret_ma_*` 9%
LightGBM	MDI / gain	`har_ma_*` 57%; `sumret_ma_*` 5%; `sumabsret_ma_*` 4%
LightGBM	split count	`har_ma_*` 12%; `sumret_ma_*` 6%; `sumvolume_ma_*` 4%
LightGBM	perm. P1, ΔQLIKE	`har_ma_*` +111%; `calendar` +5%; `sumret_ma_*` +4%
LightGBM	perm. P2, ΔQLIKE	`har_ma_*` +237%; `calendar` +6%; `sumret_ma_*` +4%
LightGBM	perm. P2, ΔMSE	`har_ma_*` +235%; `calendar` +4%; `sumret_ma_*` +3%
LightGBM	SHAP mean $\|\phi\|$	`har_ma_*` 47%; `sumret_ma_*` 6%; `sumvolume_ma_*` 4%
XGBoost	MDI / gain	`har_ma_*` 69%; `sumret_ma_*` 4%; `sumabsret_ma_*` 4%
XGBoost	split count	`har_ma_*` 21%; `sumret_ma_*` 8%; `sumvolume_ma_*` 5%
XGBoost	perm. P1, ΔQLIKE	`har_ma_*` +107%; `sumret_ma_*` +4%; `calendar` +4%
XGBoost	perm. P2, ΔQLIKE	`har_ma_*` +224%; `sumret_ma_*` +4%; `calendar` +4%
XGBoost	perm. P2, ΔMSE	`har_ma_*` +231%; `sumret_ma_*` +4%; `sumvolume_ma_*` +3%
XGBoost	SHAP mean $\|\phi\|$	`har_ma_*` 55%; `sumret_ma_*` 7%; `sumvolume_ma_*` 5%
random forest	MDI / gain	`har_ma_*` 62%; `sumabsret_ma_*` 5%; `sumret_ma_*` 3%
random forest	split count	`har_ma_*` 4%; `voldemand_all_open_and_close_ma_*` 3%; `voldemand_spx_open_and_close_ma_*` 3%
random forest	perm. P1, ΔQLIKE	`har_ma_*` +128%; `calendar` +2%; `sumret_ma_*` +2%
random forest	perm. P2, ΔQLIKE	`har_ma_*` +290%; `calendar` +3%; `sumabsret_ma_*` +3%
random forest	perm. P2, ΔMSE	`har_ma_*` +252%; `calendar` +1%; `sumvolume_ma_*` +1%
random forest	SHAP mean $\|\phi\|$	`har_ma_*` 58%; `sumabsret_ma_*` 5%; `sumret_ma_*` 4%

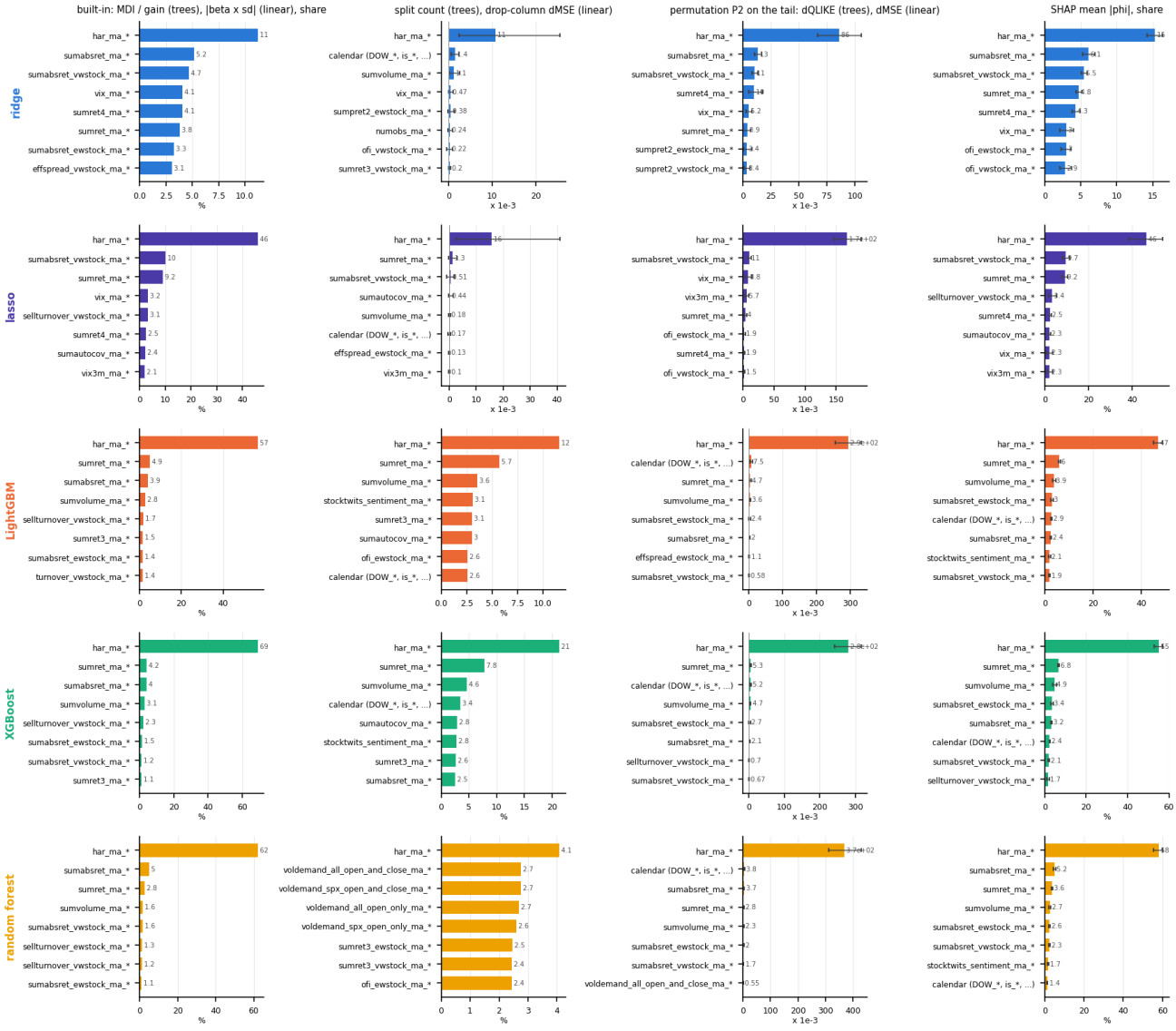


Figure 1: All features: ranked bars per measure (series level; top 8). Rows: models; columns: the built-in measure, split count (trees) / drop-column (linear), permutation P2 on the tail, SHAP. Bars with 95% intervals over refits. Linear permutation / drop-column in ΔMSE (see Section 6).

3. Ranking agreement

Table 5: Within a tree model: Spearman between measures (series / column level).

bucket	level	model	MDI∼split	MDI∼SHAP	MDI∼P2	SHAP∼P2	P1∼P2	P2 QL∼MSE
all features	series	LightGBM	0.87	0.87	0.22	0.25	0.75	0.70
all features	series	XGBoost	0.87	0.85	0.54	0.57	0.80	0.57
all features	series	random forest	0.64	0.86	0.22	0.30	0.67	0.57
all features	column	LightGBM	0.99	0.97	0.06	0.03	0.51	0.56
all features	column	XGBoost	0.99	0.97	0.29	0.28	0.60	0.45
all features	column	random forest	0.97	0.98	0.12	0.13	0.58	0.56

Table 6: Within a linear model: Spearman between measures.

bucket	level	model	$\lvert\beta sd\rvert$ ∼SHAP	$\lvert\beta sd\rvert$ ∼P2	SHAP∼P2	P1∼P2	P2∼drop	P2 QL∼MSE
all features	series	ridge	0.83	0.60	0.74	0.76	0.36	0.06
all features	series	lasso	0.95	0.88	0.86	0.83	0.16	0.96
all features	column	ridge	0.99	0.40	0.39	0.59	0.22	0.04
all features	column	lasso	0.99	0.32	0.31	0.61	0.10	0.78

Table 7: Across models, same measure: range of Spearman over tree–tree pairs, tree–linear pairs, and ridge–lasso.

bucket	level	SHAP			perm P2 ΔQLIKE			perm P2 ΔMSE		
		tree	tree–lin	r–l	tree	tree–lin	r–l	tree	tree–lin	r–l
all features	series	0.83–0.89	0.33–0.49	0.86	0.43–0.59	-0.22–0.39	-0.04	0.30–0.62	0.04–0.45	0.72
all features	column	0.92–0.97	0.65–0.83	0.78	0.19–0.36	-0.18–0.10	-0.00	0.22–0.32	0.01–0.12	0.41

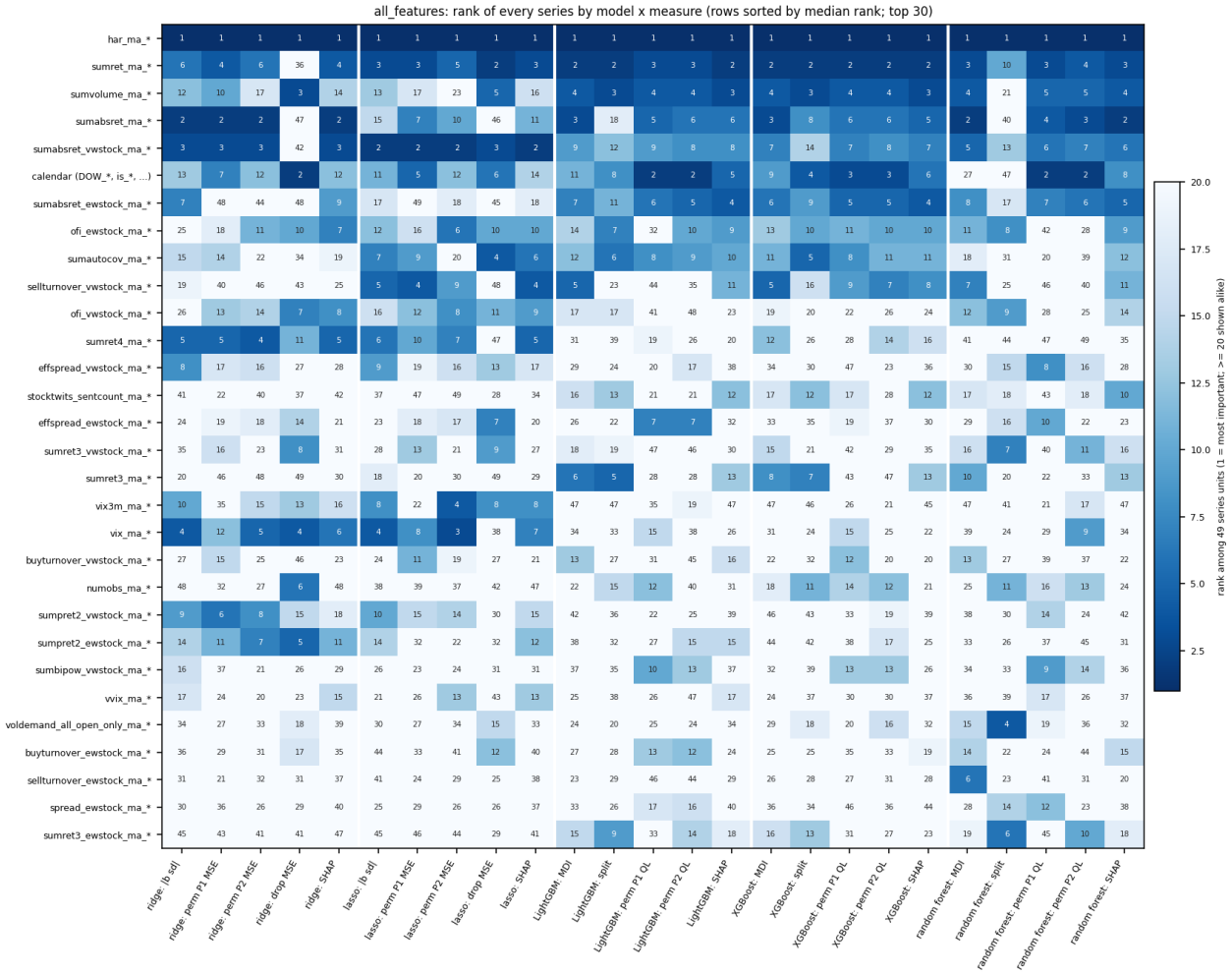


Figure 2: All features: rank of every series under every model × measure (1 = most important).

4. Stability over refits and regimes

Table 8: Top-5 stability (series level): share of the 147 refits in which `har_ma_*` is in the refit's top 5; number of series in the top 5 in at least half the refits; number ever in the top 5. Measures: built-in (MDI or $|\beta sd|$), permutation P2 (ΔQLIKE trees, ΔMSE linear), SHAP.

bucket	model	built-in			perm P2			SHAP		
		har	≥50%	ever	har	≥50%	ever	har	≥50%	ever
all features	ridge	100%	5	11	95%	3	40	99%	3	27
all features	lasso	100%	4	12	100%	3	31	100%	3	24
all features	LightGBM	100%	4	11	98%	2	48	100%	3	34
all features	XGBoost	100%	5	10	99%	3	45	100%	4	18
all features	random forest	100%	4	13	99%	1	47	100%	5	25

Table 9: Regimes: Spearman between the ranking within a stratum and the full-sample ranking (series level): median (minimum) over the calendar years (2018: 13, 2019: 26, 2020: 25, 2021: 25, 2022: 25, 2023: 25, 2024: 8 refits) and over the causal VIX terciles (VIX at 15:30 against the 1/3, 2/3 quantiles of the previous 2000 sessions' 15:30 VIX: low 28, mid 38, high 81 refits); years in which the full-sample leader also leads.

bucket	model	built-in			perm P2			SHAP		
		year	VIX	lead	year	VIX	lead	year	VIX	lead
all features	ridge	0.93 (0.84)	0.99 (0.95)	7/7	0.69 (0.49)	0.82 (0.58)	7/7	0.76 (0.54)	0.92 (0.83)	7/7
all features	lasso	0.61 (0.47)	0.99 (0.85)	7/7	0.41 (0.14)	0.75 (0.34)	7/7	0.64 (0.43)	0.97 (0.83)	7/7
all features	LightGBM	0.95 (0.86)	1.00 (0.99)	7/7	0.56 (0.22)	0.70 (0.67)	7/7	0.83 (0.72)	0.93 (0.92)	7/7
all features	XGBoost	0.93 (0.88)	0.99 (0.97)	7/7	0.61 (0.10)	0.59 (0.55)	7/7	0.83 (0.67)	0.91 (0.91)	7/7
all features	random forest	0.98 (0.95)	1.00 (0.99)	7/7	0.55 (0.36)	0.64 (0.42)	7/7	0.90 (0.78)	0.97 (0.96)	7/7

5. Correlated inputs: the dilution caveat

Table 10: Joint permutation of a cluster against the sum of its columns permuted one at a time (P2, ΔMSE in the fit space $\times 10^3$, joint / sum; squared error so the linear columns are not dominated by single tails), for the four clusters with the largest joint effect in LightGBM. Joint > sum: the columns substitute for one another and one-at-a-time permutation understates the group (dilution). Joint < sum: offsetting weights (permuting one column breaks an offset its partner carries).

bucket	cluster	members	ridge	lasso	LightGBM	XGBoost	forest
all features	C49	`har_ma_1, har_ma_5, har_ma_1_x_close, har_ma_5_x_close` (4; $\|r\| \geq 0.89$)	33.2 / 17.6	85.5 / 48.9	101.3 / 32.0	96.0 / 35.4	136.1 / 29.0
all features	C46	`har_ma_25, har_ma_125, har_ma_25_x_close, har_ma_125_x_close` (4; $\|r\| \geq 0.83$)	13.4 / 7.3	19.9 / 11.8	6.3 / 2.9	7.0 / 3.5	3.0 / 1.5
all features	C50	`sumabsret_ewstock_ma_1, sumabsret_ewstock_ma_5, sumabsret_vwstock_ma_1, sumabsret_vwstock_ma_5` (4; $\|r\| \geq 0.85$)	15.7 / 3.0	8.8 / 6.5	1.2 / 0.2	0.8 / -0.3	1.4 / 0.1
all features	C48	`sumvolume_ma_1, sumvolume_ma_5` (2; $\|r\| \geq 0.89$)	1.1 / 1.4	0.5 / 0.6	0.8 / 0.5	1.3 / 1.0	0.5 / 0.5

6. Linear models: QLIKE permutation is dominated by single tails

The linear forecasts are unbounded: moving an extreme input (a crash day's 4th-power or absolute return) onto another row can push the fit-space forecast through zero, and QLIKE of $\hat{y}^2\times$ baseline explodes. Every permutation / drop-column table therefore carries the median refit and the largest single-refit share of the sum, and the linear figures and cross-model comparisons use the squared-error versions. Trees cannot extrapolate beyond the training range of the target, so their QLIKE changes are well behaved.

Table 11: The top linear QLIKE entries: mean [95% interval], median refit, share of the sum carried by one refit.

bucket	model	measure	top series	mean [interval]	median	one refit
all features	ridge	perm. P1, ΔQLIKE	sumret4_ewstock_ma_*	315 [-0.0012, 9.4e+02]	6.1e-05	100%
all features	ridge	perm. P2, ΔQLIKE	sumautocov_ma_*	1.03e+05 [-0.012, 3.1e+05]	0.0022	100%
all features	ridge	drop-column, ΔQLIKE	sumabsret_ma_*	980 [0.0016, 2.9e+03]	-0.00025	100%
all features	lasso	perm. P1, ΔQLIKE	har_ma_*	0.144 [0.12, 0.17]	0.11	7%
all features	lasso	perm. P2, ΔQLIKE	har_ma_*	0.322 [0.28, 0.36]	0.24	4%
all features	lasso	drop-column, ΔQLIKE	har_ma_*	38.7 [0.0065, 1.2e+02]	0.0052	100%

7. Tuned trees

Table 12: Causally tuned trees (their own extracts: gain per refit and TreeSHAP rows at 16:00): har_ma_* share tuned (untuned), rank correlation tuned vs untuned over the series, and the next three series by tuned SHAP. Split count and permutation were not computed for the tuned trees (their per-refit configurations would have to be refitted).

bucket	model	MDI har	SHAP har	ρ MDI	ρ SHAP	next by tuned SHAP
all features	LightGBM	60% (57%)	46% (47%)	0.92	0.91	sumret, sumvolume, sumabsret
all features	XGBoost	54% (69%)	41% (55%)	0.96	0.91	sumret, sumvolume, sumabsret
all features	random forest	51% (62%)	48% (58%)	0.79	0.78	sumabsret, sumvolume, sumabsret_vwstock

8. Gates

- Design: out-of-sample stamps, targets and column names equal the stored tree runs'; the linear captures equal the design rows exactly and the stored research forecasts to $\leq$ 7e-11 (relative).
- Refits: XGBoost and random forest reproduce the stored forecasts (largest gap 1.3e-15; above 10^{-9} on 0 XGBoost and 0 forest refits of the 2×147). LightGBM reproduces them exactly on HAR + calendar (gap 4.4e-16) but not on all features: mean |gap| 0.0093, max 0.058, correlation 0.9995 (same version, parameters, seed and thread count; deterministic run to run on two machines; XGBoost reproduces on the same input). The LightGBM numbers are those of the refit.
- Drop-column anchors reproduce the captured coefficients (ridge $\leq$ 2e-10, lasso $\leq$ 5e-04). Every model of a bucket saw identical permutation draws (md5 per refit). TreeSHAP additivity $\leq$ 3e-06.

Files: experiments/feature_importance_1530.py (design, inputs, linear, aggregate); experiments/feature_importance_1530_trees.py (the refits, run on the cluster through cluster/slurm/submit_featimp.sh); tables and figures in results/feature_importance_1530/; summary SUMMARY.md.

Appendix: HAR + calendar design

Table 13: HAR + calendar, series level: top three per model and measure.

model	measure	top three
ridge	$\lvert\beta \cdot sd\rvert$	`har_ma_*` 81%; `calendar` 19%
ridge	perm. P1, ΔMSE	`har_ma_*` +280%; `calendar` +8%
ridge	perm. P2, ΔMSE	`har_ma_*` +501%; `calendar` +9%
ridge	perm. P2, ΔQLIKE	`har_ma_*` +537%; `calendar` +10%
ridge	drop-column, ΔMSE	`har_ma_*` +247%; `calendar` +5%
ridge	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 90%; `calendar` 10%
lasso	$\lvert\beta \cdot sd\rvert$	`har_ma_*` 84%; `calendar` 16%
lasso	perm. P1, ΔMSE	`har_ma_*` +276%; `calendar` +7%
lasso	perm. P2, ΔMSE	`har_ma_*` +495%; `calendar` +8%
lasso	perm. P2, ΔQLIKE	`har_ma_*` +539%; `calendar` +9%
lasso	drop-column, ΔMSE	`har_ma_*` +244%; `calendar` +4%
lasso	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 91%; `calendar` 9%
LightGBM	MDI / gain	`har_ma_*` 96%; `calendar` 4%
LightGBM	split count	`har_ma_*` 87%; `calendar` 13%
LightGBM	perm. P1, ΔQLIKE	`har_ma_*` +225%; `calendar` +8%
LightGBM	perm. P2, ΔQLIKE	`har_ma_*` +483%; `calendar` +8%
LightGBM	perm. P2, ΔMSE	`har_ma_*` +426%; `calendar` +6%
LightGBM	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 91%; `calendar` 9%
XGBoost	MDI / gain	`har_ma_*` 97%; `calendar` 3%
XGBoost	split count	`har_ma_*` 82%; `calendar` 18%
XGBoost	perm. P1, ΔQLIKE	`har_ma_*` +219%; `calendar` +7%
XGBoost	perm. P2, ΔQLIKE	`har_ma_*` +457%; `calendar` +7%
XGBoost	perm. P2, ΔMSE	`har_ma_*` +444%; `calendar` +5%
XGBoost	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 93%; `calendar` 7%
random forest	MDI / gain	`har_ma_*` 95%; `calendar` 5%
random forest	split count	`har_ma_*` 82%; `calendar` 18%
random forest	perm. P1, ΔQLIKE	`har_ma_*` +232%; `calendar` +7%
random forest	perm. P2, ΔQLIKE	`har_ma_*` +528%; `calendar` +7%
random forest	perm. P2, ΔMSE	`har_ma_*` +453%; `calendar` +5%
random forest	SHAP mean $\lvert\phi\rvert$	`har_ma_*` 93%; `calendar` 7%

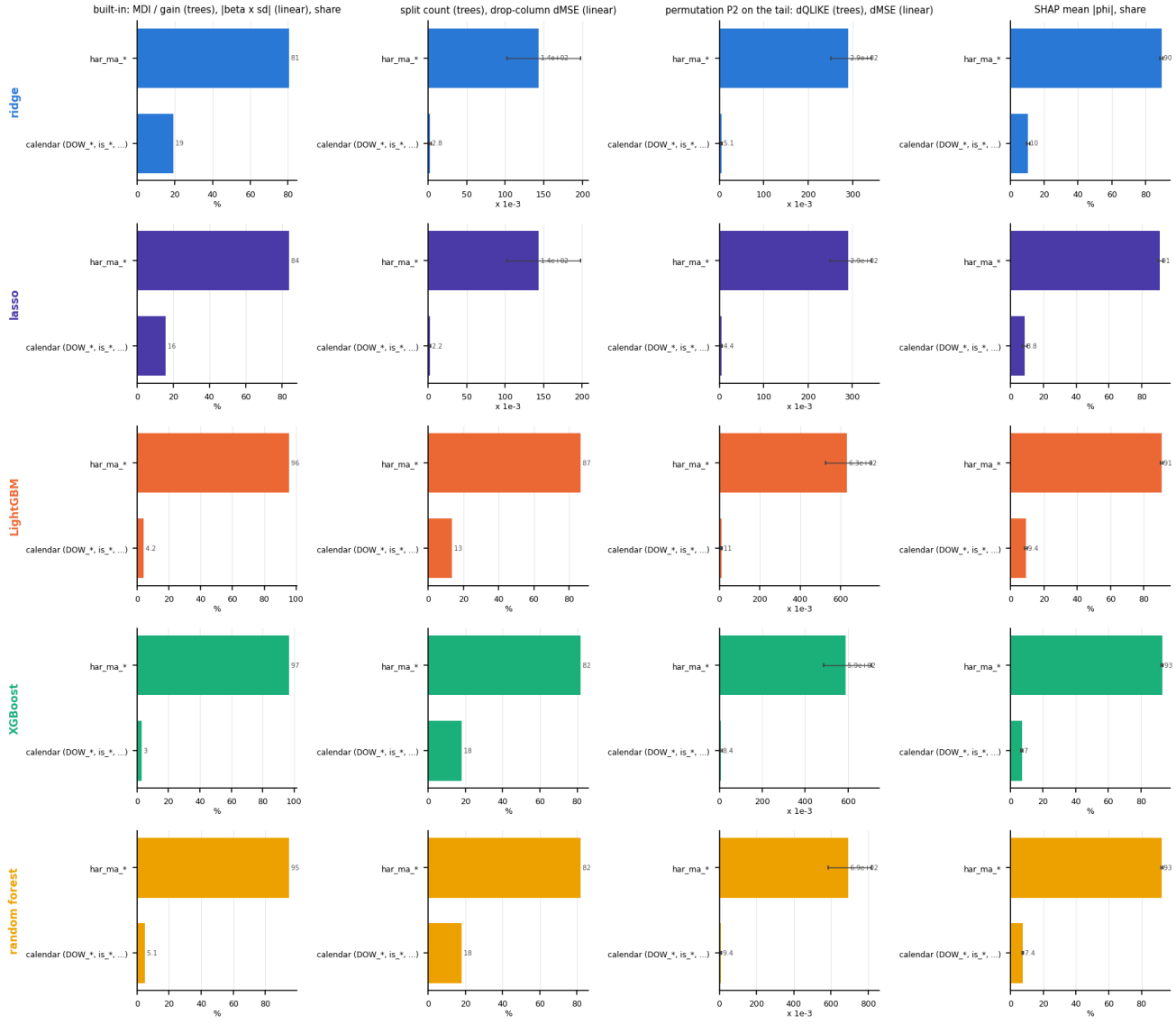


Figure 3: HAR + calendar: ranked bars per measure.

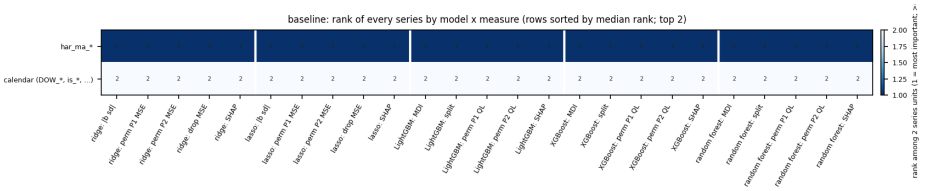


Figure 4: HAR + calendar: ranks across measures.

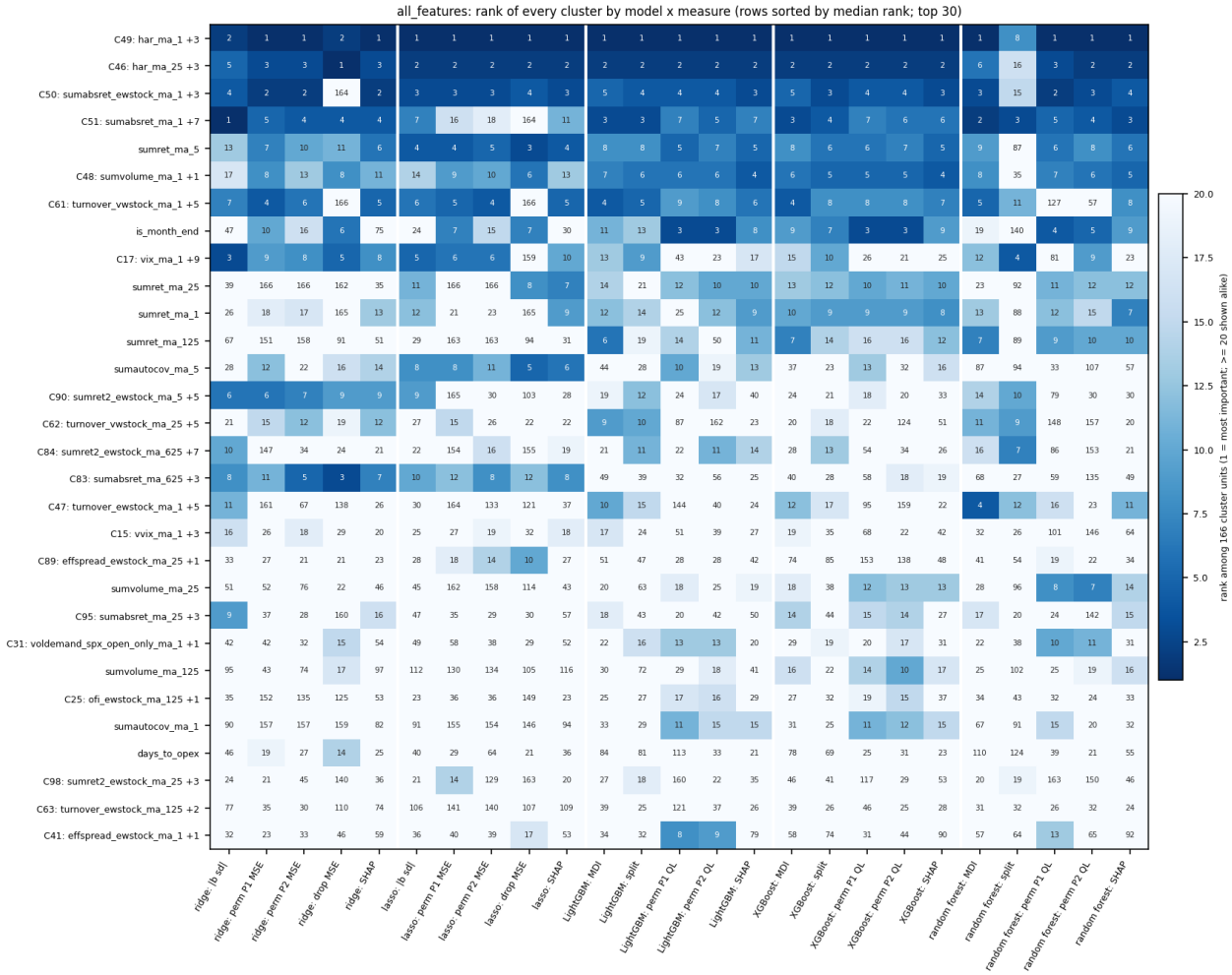


Figure 5: All features: ranks at the cluster level.